PRACTICAL 10

DATA PREPARATION AND TRANSFORMATION USING TABLEAU PREP

MAIN PURPOSE OF THIS PRACTICAL:-

We are learning how to prepare raw or messy data before using it for analysis and visualization in Tableau.

Real-world data is usually not ready to use directly. It may contain:

- Missing values
- Duplicate or inconsistent data
- Data in the wrong format
- Data spread across multiple files
- Data arranged in a difficult format
- Data from different databases

So, in this practical, we learn how to clean, combine, organize, and restructure the data so that Tableau can create correct and meaningful charts, dashboards, and analysis.

AIM:- To prepare, clean, transform, and organize data using Tableau Prep for analysis and visualization.

DATASET TO USE

We will use a simple Employee Department Dataset.

You can create an Excel file with the following data:

Employee_ID	Employee_Name	Department	City	Salary	Joining_Year
E101	Amit	IT	Mumbai	45000	2022
E102	Neha	it	Mumbai	50000	2021
E103	Rahul	IT Dept	Thane	48000	2023
E104	Priya	HR	Mumbai	40000	2022
E105	Sneha	Human Resource	Thane	42000	2021
E106	Karan	Sales	Palghar	38000	2023
E107	Riya	sales	Palghar	41000	2022
E108	Ajay	IT	Nashik	52000	2020
E109	Pooja	HR	Nashik	43000	2023
E110	Vijay	IT Dept	Mumbai	55000	2021

This dataset is deliberately made with inconsistent values so that students can practice cleaning.

For example:

- IT
- it
- IT Dept

actually refer to the same department.

Similarly:

- HR
- Human Resource

refer to the same department.

A) CONNECT AND LOAD DATA INTO TABLEAU PREP

STEP 1: Open Tableau Prep

Open:

Tableau Prep Builder

You will see the start screen.

STEP 2: Connect the Excel file

On the left side, under Connections:

Click:

Microsoft Excel

Select your Excel file.

For example:

Employee_Data.xlsx

STEP 3: Select the sheet

Tableau Prep will show the sheets in your Excel file.

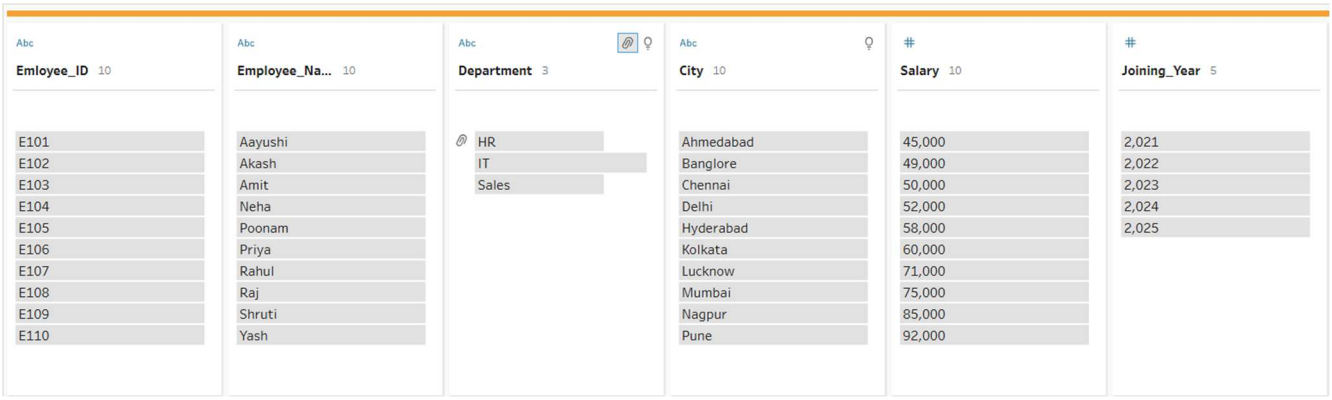
Select:

Employee Data

Drag it to the flow area.

You will see:

Input



Emloyee_ID 10	Employee_Na... 10	Department 3	City 10	Salary 10	Joining_Year 5
E101	Aayushi	HR	Ahmedabad	45,000	2,021
E102	Akash	IT	Banglore	49,000	2,022
E103	Amit	Sales	Chennai	50,000	2,023
E104	Neha		Delhi	52,000	2,024
E105	Poonam		Hyderabad	58,000	2,025
E106	Priya		Kolkata	60,000	
E107	Rahul		Lucknow	71,000	
E108	Raj		Mumbai	75,000	
E109	Shruti		Nagpur	85,000	
E110	Yash		Pune	92,000	

STEP 4: Add a Cleaning Step

Click the + symbol after Input.

Select:

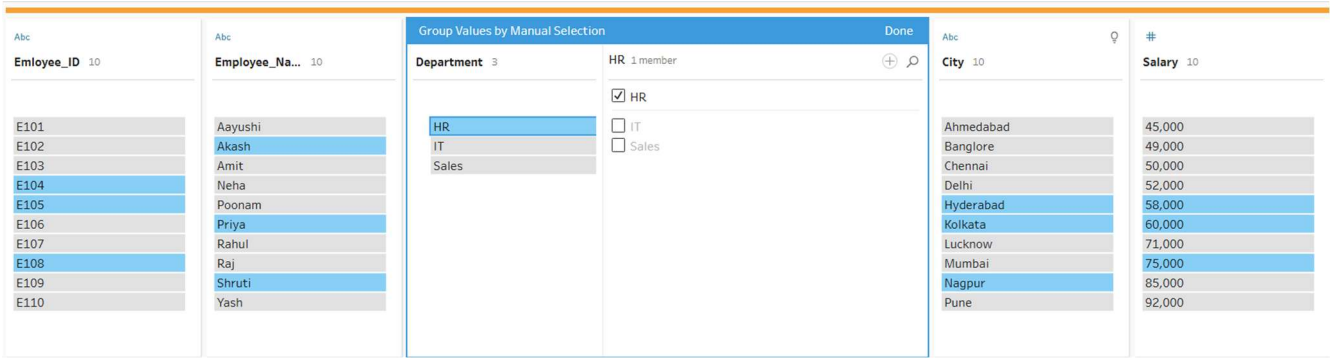
Clean Step

Your flow will look like:

Input → Clean

This is where we will clean the data.

B) CLEAN AND PROFILE DATA



Emloyee_ID 10	Employee_Na... 10	Department 3	City 10	Salary 10
E101	Aayushi	HR	Ahmedabad	45,000
E102	Akash	IT	Banglore	49,000
E103	Amit	Sales	Chennai	50,000
E104	Neha		Delhi	52,000
E105	Poonam		Hyderabad	58,000
E106	Priya		Kolkata	60,000
E107	Rahul		Lucknow	71,000
E108	Raj		Mumbai	75,000
E109	Shruti		Nagpur	85,000
E110	Yash		Pune	92,000

What does cleaning mean?

Cleaning means:

Finding and correcting problems in the data.

Examples:

- Missing values
- Spelling mistakes
- Extra spaces
- Different formats
- Duplicate records
- Incorrect values

STEP 1: Examine the data

Click the Clean Step.

Tableau Prep displays the fields.

You can see:

- Employee_ID
- Employee_Name
- Department
- City
- Salary
- Joining_Year

Tableau Prep also shows the distribution of values.

This is called data profiling.

Simple meaning:

Data profiling = checking what is inside our data.

STEP 2: Check Department

Look at the Department field.

You will notice:

IT

it

IT Dept

HR

Human Resource

Sales

sales

There are inconsistencies.

We will fix them in the next step.

C) GROUP AND STANDARDIZE VALUES

This is one of the most important parts of Tableau Prep.

What does standardization mean?

It means:

Changing different versions of the same value into one common value.

For example:

IT

it

IT Dept

becomes:

IT

Do this:

1. In your Clean Step, find Department.
2. Click the drop-down/menu on the Department field.
3. Select Group Values.
4. Tableau Prep will show the different values:

- IT
- it
- IT Dept
- HR
- Human Resource
- Sales
- sales

5. Select IT, it, and IT Dept.
6. Click Group.

They will become one group.

7. Rename the group to IT.

Then do the same for:

HR + Human Resource → HR

and

Sales + sales → Sales

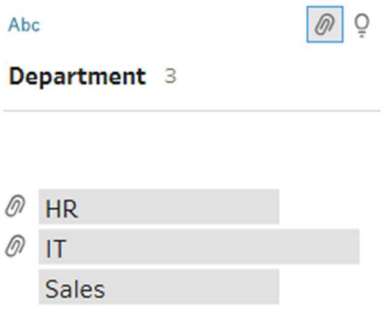
Your final Department values should be:

Before	After
IT	IT
it	IT
IT Dept	IT
HR	HR
Human Resource	HR
Sales	Sales
sales	Sales

So, just click “Group Values” — you are on the right option. ✅

If you want, we can continue from this exact step to sub-practical (d) Merge Mismatched Fields, one clicks at a time.

Expected Output:-



D) MERGE MISMATCHED FIELDS

For this part, we will use two files and combine them using a common field.

Our first file:

Employee_Data.xlsx

contains:

Employee_ID	Employee_Name	Department	City	Salary
E101	Amit	IT	Mumbai	45000
E102	Neha	IT	Mumbai	50000

E103	Rahul	IT	Thane	48000

Second file:

Create another Excel file called Employee_Location.xlsx:

Employee_ID	Location
E101	Mumbai Office
E102	Mumbai Office
E103	Mumbai Office
E104	Mumbai Office
E105	Thane Office

The common field is:

Employee_ID

STEP 1: Add the second file

In Tableau Prep:

8. Go to the Connections area on the left.
9. Click +.
10. Select Microsoft Excel.
11. Select Employee_Location.xlsx.
12. Select the sheet.
13. Drag the sheet into the flow.

You should now have two inputs:

Employee Data

and

Employee Location

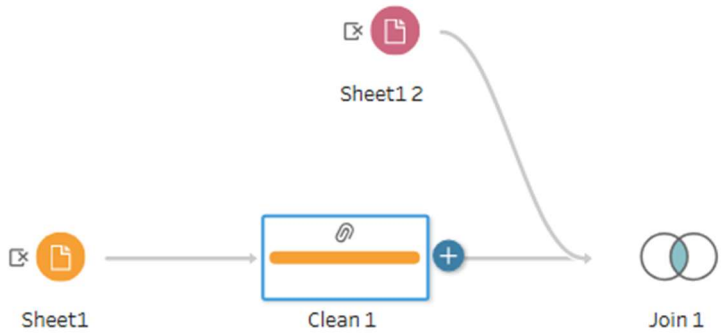
STEP 2: Create a Join

We want to combine these two tables.

14. Click the + after your Employee Data step.
15. Select Join.
16. Drag/connect the second table to the Join step.

You should see something like:

Employee Data → Join ← Employee Location



STEP 3: Select the matching field

Tableau Prep needs to know:

Which column is common in both files?

Select:

Employee_ID = Employee_ID

This is called the Join Clause.

STEP 4: Check the joined data

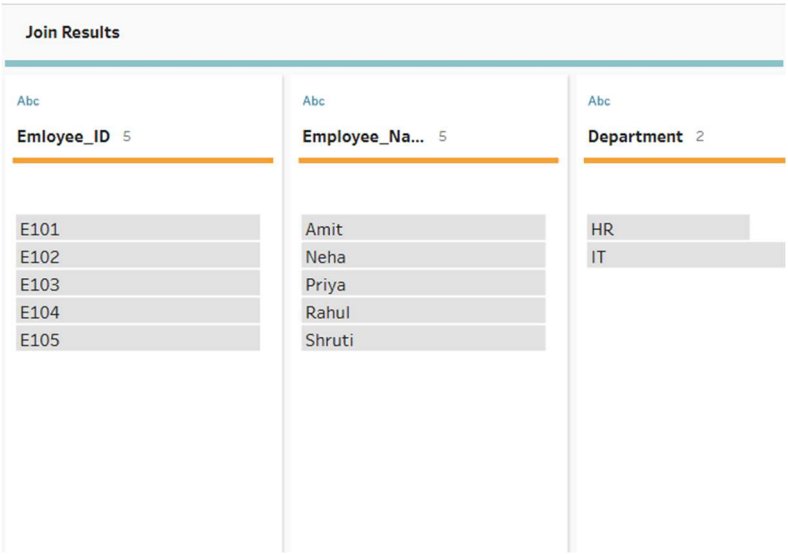
Click the Join step.

You should now see fields from both files.

For example:

Employee_ID	Employee_Name	Department	Salary	Location
E101	Amit	IT	45000	Mumbai Office
E102	Neha	IT	50000	Mumbai Office
E103	Rahul	IT	48000	Thane Office

Now the two files have been combined.



SIMPLE MEANING

Join = combining two tables using a common field.

Here:

Employee_ID is the common field.

(After joining first count filter and clean data and after that count aggregate function.)

E) PERFORM AGGREGATION

Now let's continue.

STEP 1: Add Aggregate

17. Click the + after the Join step.
18. Select Aggregate.



STEP 2: Group by Department

In the Aggregate step:

Drag:

Department → Grouped Fields

This means:

"Give me a separate result for each department."

STEP 3: Calculate Average Salary

Find:

Salary

Drag it into Aggregated Fields.

Change the aggregation to:

Average

STEP 4: Count Employees

Drag:

Employee_ID

to Aggregated Fields.

Set it to:

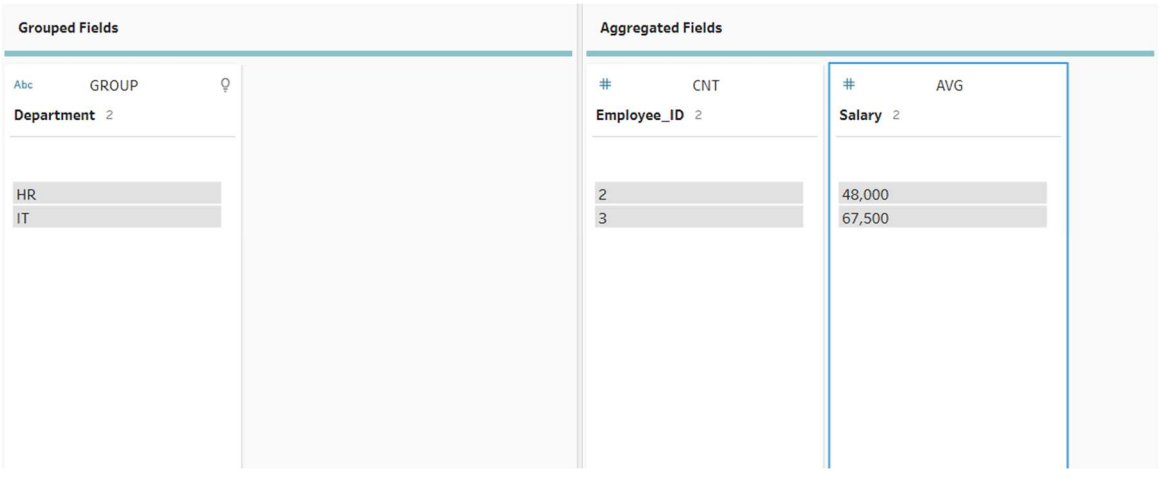
Count

Your result will look approximately like:

Department	Count of Employee_ID	Average Salary
IT	4	50,000
HR	2	41,500
Sales	2	39,500

The exact numbers depend on the complete dataset.

Expected Output:-



F) APPLY FILTERS AND TRANSFORMATIONS

Now let's move to the next sub-practical.

STEP 1: Add a Clean Step

Click the + after the Aggregate step.

Select:

Clean Step

You should now have:

Join → Aggregate → Clean

STEP 2: Filter the data

Suppose we want only employees whose salary is more than ₹40,000.

In the Clean step:

19. Find the Salary field.
20. Click its drop-down/menu.
21. Look for Filter.
22. Set the condition:

Salary > 40000

23. Apply the filter.

Now Tableau Prep will keep only records where salary is greater than ₹40,000.

Open the field menu for Joining_Year. how to open it

In Tableau Prep, the field menu is the small menu attached to the field card.

For Joining_Year:

24. Open your Clean Step.
25. Find the field card named Joining_Year.
26. Move your mouse over the Joining_Year field.
27. Look for a small down-arrow (▼) or three-dot menu (⋯) on the field card.
28. Click it.
29. A menu will open.
30. Choose Rename Field.
31. Type:

Year

32. Press Enter.

If you don't see Joining_Year

It may have disappeared because you are currently looking at the Aggregate step. Click the Clean Step before the Aggregate where all the original fields are visible.

If your screen looks different, send me a screenshot of your Clean Step and I can point out exactly where the menu is.

there is no card name like joining year

For Practical 10, do this instead:

Since you have already completed:

Input → Clean → Group Values → Join → Aggregate

we can skip the Joining_Year rename step.

For the Filter and Transformation sub-practical, use a field that you can actually see.

Let's filter Salary

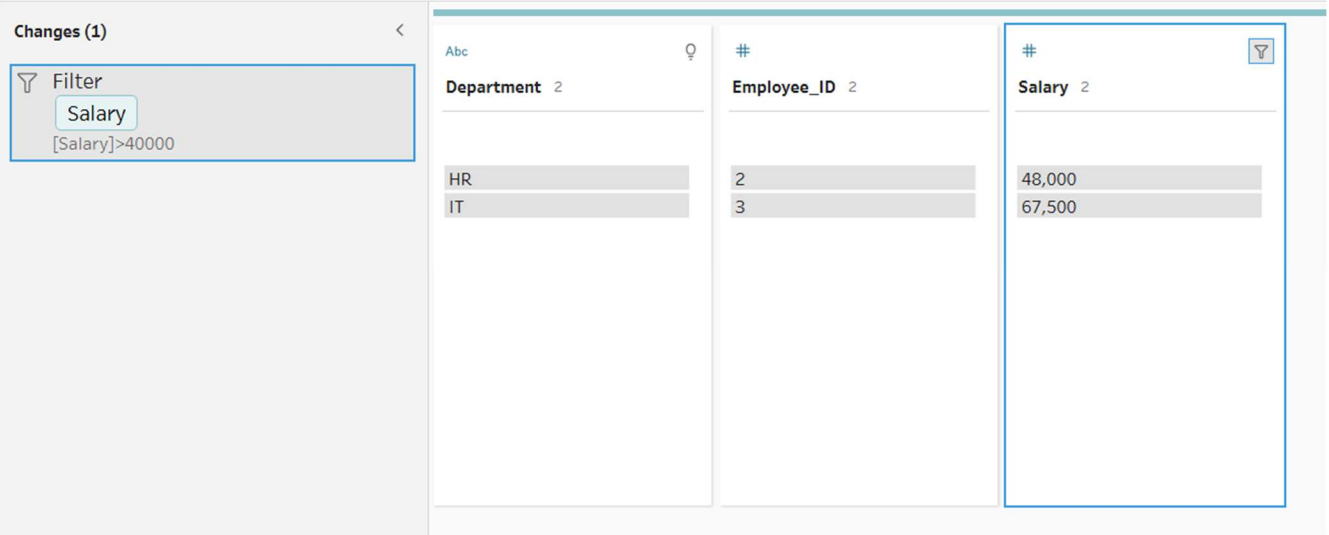
33. Click your Clean Step after the Join/Aggregate step.
34. Look at the fields displayed.
35. Find Salary.
36. Click the small drop-down arrow / menu on the Salary field.
37. Choose the filtering option.
38. Set:

Salary > 40000

39. Apply the filter.

That's enough to demonstrate filtering.

Expected Output:-



Conclusion:-

In this practical, we learned how to prepare, clean, transform, and organize raw data using Tableau Prep. We identified and handled missing, duplicate, and inconsistent values to improve data quality. We also learned how to change data formats, combine data from different sources, and restructure data into a suitable format. Tableau Prep makes the data preparation process easier and more efficient before analysis. The cleaned and transformed data can be used to create accurate charts, dashboards, and meaningful visualizations in Tableau. Overall, this practical helped us understand the importance of data preparation in obtaining reliable and meaningful results from data analysis.